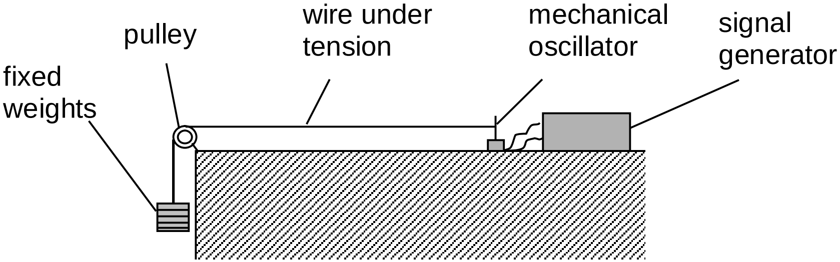
Similarly, the graph in Fig. 6.2 for cooking oil should scale down along the $Q$-axis.

**7**

Diagram:




Aim: To determine the value of $n$ from the gradient of the graph of $\lg f$ against $\lg \mu$.

Independent variable: $\mu$, mass per unit length of the wire.

The length $L$ of the wire can be measured by using a metre rule and its mass $m$ can be determined using the weighing balance. Hence, $\mu$ can be calculated using $\mu = \frac{m}{L}$.

Dependent variable: $f$, frequency of the mode of vibration of the wire.

The frequency can be recorded from the signal generator.

Control variables:

- same length of wire, measured using metre rule
- same tension in the wire, by ensuring same weights hung
- same mode of vibration, by adjusting the frequency until the $1^{st}$ overtone is observed (Fig. 7.1).

Procedure:

(a) Measure and record down the mass $m$ of a wire of fixed length $L$ (about 0.80 m) using the weighing balance.

(b) Set up the apparatus as shown in the diagram by attaching the wire to the fixed weights.

(c) Increase the frequency $f$ of the signal generator from zero until a standing wave ($1^{st}$ overtone) is observed. measure using a light gates connected to a data logger to the frequency $f$ shown on the on signal generator in the table.

(d) Adjust the frequency of the oscillator back to zero and repeat the experiment to obtain 3 values of $f$. Calculate the average value of frequency, $f_{average}$.

(e) Replace the wire with another one of different mass but same length $L$. Conduct the experiment from steps (a) to (d) for at least 6 different wires.

(f) Calculate the values of mass per unit length $\mu$, where $\mu = \frac{m}{L}$ for each of the 6 wires used.

(g) On the same table, calculate the values of $\lg \mu$ and $\lg f$.

(f) Plot a graph of $\lg f$ against $\lg \mu$. Determine the value of $n$ form the gradient of the graph since $\lg f = n \lg \mu + \lg k$.

Precaution: Ensure that the length of the wires used is the same throughout and the length between the pulley and oscillator is the same throughout.

Safety: Clamp the oscillator to the bench using a g-clamp. Also, avoid standing too close to the set-up while adjusting the frequency as the wire might dislodge from the oscillator.